Supplementary Material for "A Semantic Mutation Metric for Metamorphic-Relation Adequacy for Scientific-Computing Kernels"

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CCS Concepts: • **Software and its engineering → Software testing and debugging**; *Software verification and validation.*

Additional Key Words and Phrases: metamorphic testing, mutation testing, semantic mutation testing, metamorphic relation adequacy, metamorphic relation assessment, scientific computing

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A Notation and Operator Catalogue

A.1 Complete notation table

Experimental objects.

- PUT set $I = \{\text{A1,A2,A3, B1,B2,B3, C1,C2,C3, D1,D2,D3}\}$, $|I| = 12$.
- PUT: S_i ($i \in I$).
- Class mapping: $\text{cls} : I \to \{A, B, C, D\}$ (open, extensible).
- Valid input distribution: D_S, sampling $X_{K_{\text{eq}}} \sim D_S$.

Metamorphic testing side.

- Metamorphic-relation label set: $\mathcal{R} = \{\text{Cons., Mono., Conv., Traj., P-ord.}\}$, $k \in \{1, \ldots, 5\}$ (open, extensible).
- Metamorphic Relation: $\text{MR}_{i,k}$ (from the shared infrastructure); $mr = (r, R) \in \text{MR}_{i,k}$.
- Automated Verification Pipeline (AVP): $\text{AVP} : \text{Programs} \times \text{MR}_{\text{universe}} \times \mathbb{R}^+ \to \{\text{pass, fail}\}$; $\text{AVP}(s, mr, \varepsilon^k_{\text{AVP}})$ invokes the verification method per relation index k.

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Mutation testing side.

- Mutation operator family: $\mathrm{MUT} = \{\mathrm{mut}_1, \ldots, \mathrm{mut}_5\}$, $j \in \{1, \ldots, 5\}$ (open, extensible); $\mathrm{mut}_1 = \mathrm{mut}_C$, $\mathrm{mut}_2 = \mathrm{mut}_M$, $\mathrm{mut}_3 = \mathrm{mut}_G$, $\mathrm{mut}_4 = \mathrm{mut}_T$, $\mathrm{mut}_5 = \mathrm{mut}_F$.
- Signature: $\mathrm{mut}_j : \mathrm{Programs} \to 2^{\mathrm{Programs}}$.
- Mutant: $s' \in \mathrm{mut}_j(S_i)$.
- Alignment: $\mathrm{align}(j) = j$.

Three-state decomposition (fixed).

$$\mathrm{mut}_j(S_i) = \mathrm{equiv}_{i,k,j} \cup \mathrm{killed}_{i,k,j} \cup \mathrm{survive}_{i,k,j},$$

with the three sets disjoint, mutually exclusive, and exhaustive.

- *equiv determination* (dual conditions): (E1) AVP-coherent: $\forall mr \in \mathrm{MR}_{i,k} : \mathrm{AVP}(S_i, mr) = \mathrm{AVP}(s', mr)$; (E2) Output-equiv: $\forall x \in X_{K_{\mathrm{eq}}} \sim D_S : \|S_i(x) - s'(x)\| \le \varepsilon_{\mathrm{eq}}$.
- *killed determination*: $\mathrm{killed}(s', \mathrm{MR}_{i,k}) \Leftrightarrow \exists mr \in \mathrm{MR}_{i,k} : \mathrm{AVP}(S_i, mr) = \mathrm{pass} \wedge \mathrm{AVP}(s', mr) = \mathrm{fail}$.

Metric (fixed classical structure).

$$\begin{aligned}\mathrm{SMS}_{i,k,j} &:= \frac{|\mathrm{killed}_{i,k,j}|}{|\mathrm{mut}_j(S_i)| - |\mathrm{equiv}_{i,k,j}|} \\ &= \frac{|\mathrm{killed}_{i,k,j}|}{|\mathrm{killed}_{i,k,j}| + |\mathrm{survive}_{i,k,j}|} \in [0, 1].\end{aligned}$$

LRCA engineering attribution layer (descriptive, not in SMS)..

- Likely root-cause inventory: $C = \{C_1, \ldots, C_5\}$ (open, extensible). C1 True semantic failure; C2 Tolerance perturbation; C3 OOD; C4 Statistical-assumption violation; C5 Mutator artefact.
- LRCA output: each $s' \in$ killed annotated with `root_cause`$(s') \in C$.
- Descriptive quantities: $\texttt{C1\_share}_{i,k,j}$, $\texttt{suspect\_share}_{i,k,j}$.

Slicing and cross-class.

- Slicing: aligned $j = k$; cross $j \neq k$.
- Cross-class: $\Delta\mathrm{SMS}_c$, $c \in \{A, B, C, D\}$; $\mathrm{CV}(\Delta\mathrm{SMS}) := \mathrm{std}(\Delta\mathrm{SMS}_c)/|\mathrm{mean}(\Delta\mathrm{SMS}_c)|$.

Tolerance and sampling.

- Tolerance: $\varepsilon^k_{\mathrm{AVP}}$ (relation-dependent), $\varepsilon_{\mathrm{eq}}$ (equivalence output tolerance).
- Sampling: $K_{\mathrm{eq}} = 1000$ (E2 sample size), $N = 20$ (statistical replicates).

The notation skeleton (11 core concepts + three-state decomposition + SMS formula + AVP interface) is **fixed**; the *content* of MUT, $\mathcal{R}$, C, and cls is open to extension.

A.2 Root-Cause Decision Tree and Diagnostic Protocol

Five likely root causes (open):

Table 1. Root-cause decision tree and three-layer diagnostic protocol.

Code	Meaning
C1	True semantic failure (what SMS seeks)
C2	Numerical tolerance perturbation
C3	Out-of-distribution (OOD)
C4	Statistical-assumption violation
C5	Mutator artefact

Three-layer diagnosis with L0 artefact pre-scan:

- **L0 artefact pre-scan.** Double-blind review following the main manuscript's mutant-pool prescreen and equivalence-detection protocol: dual-LLM cross-source review plus 20% manual sampling.
- **L1 tolerance robustness.** $N = 20$ replicates; fail ratio ≥ 0.80 considered stable.
- **L2 OOD triage.** For C/D classes, distinguish valid D_S^{valid} from OOD.
- **L3 statistical baseline.** For Wilcoxon/DTW, IID/stationarity pre-check on PUT samples.

Decision tree:

For each $s' \in \text{killed}_{i,k,j}$:

- **L1.** fail ratio $< 0.80 \Rightarrow$ C2; otherwise $\Rightarrow$ L2.
- **L2** (C/D classes). fail only in OOD $\Rightarrow$ C3; otherwise $\Rightarrow$ L3.
- **L3** (B/D classes + Wilcoxon/DTW). assumption violation $\Rightarrow$ C4; otherwise $\Rightarrow$ artefact recheck.
- **Artefact recheck.** artefact evidence $\Rightarrow$ C5; otherwise $\Rightarrow$ C1.

Multi-label priority: C5 > C4 > C3 > C2 > C1.

LRCA threshold calibration (9-grid scan; `lrca_calibration.json`). Best ood_band=0.02 with any tolerance_multiplier (3.0 / 10.0 / 30.0) gives mean_C1_share 0.200 / H4 12/60 (20.0%); ood_band=0.05 (default) or 0.10 gives 0.164 / 10/60 (16.7%). `tolerance_multiplier` has zero impact (L1 tolerance rarely triggered); `ood_band` is the only discriminator. Calibration ceiling 12/60 = 20% remains far below 80%, H4 unattainable on this dataset, not a calibration issue (inherent SNR of LLM-mutant pools).

A.3 Verification Protocol and Equivalence-Judgement Trade-Off

$$\text{AVP} : \text{Programs} \times \text{MR}_{\text{universe}} \times \mathbb{R}^+ \to \{\text{pass}, \text{fail}\}.$$

Per-relation verification: Cons. tolerance equality $|\text{LHS} - \text{RHS}| \leq \varepsilon$; Mono. / P $-$ ord. Wilcoxon signed-rank $\alpha = 0.05$; Conv. convergence-order + asymptotic residual ratio; Traj. DTW distance threshold ε_{DTW}. In NOETHER's taxonomy [1] these are the MR families $\mathrm{f}_{\text{inv.con}}$, $\mathrm{f}_{\text{mono.stat}}$, $\mathrm{f}_{\text{conv.rate}}$, $\mathrm{f}_{\text{conv.lim}}$, and $\mathrm{f}_{\text{mono.shape}}$; note that P-ord ($\mathrm{f}_{\text{conv.rate}}$, the $\mathcal{E}^*$ method-comparison refinement) and Conv ($\mathrm{f}_{\text{conv.lim}}$) are sibling families under the same MetaPattern m_{conv} (limit operator $\mathcal{L}^*$), so a P-ord instance reducing to a pure asymptotic residual-ratio rate check shares its acceptance machinery with the Conv checker while the registered stratum stays partial-order. AVP version = upstream infrastructure commit hash; our package embeds AVP source.

Three-candidate equivalence trade-off.

Table 2. Verification protocol and equivalence-judgement trade-off.

Determination	False-positive	False-negative	SMS bias
E1 alone	numerical coincidence + AVP-coherent	E1 false → non-equiv	High (false equivalents shrink the denominator)
E2 alone	output match, MR differs (rare)	K_eq miss, full-space equal	High
E1 ∧ E2	both mis-pass simultaneously	E1 ∨ E2 false → non-equiv	Slightly low (conservative; misjudged equivalents stay in the denominator as unkillable survivors)

E1 alone is fooled by insufficient AVP coverage; E2 alone by numerical coincidence on K_{eq}. E1 ∧ E2 is the conservative complete instantiation; under L_{equiv} it reduces almost-everywhere to classical bitwise equivalence (Lemma G.1). Counter-examples: E2 passes but E1 does not (rare; mutant coincides at K_eq but deviates at MR trigger points → non-equivalent); E1 passes but E2 does not (common; consistent within MR but numerical drift > $\varepsilon_{\mathrm{eq}}$ → non-equivalent).

B Experimental Subjects and Mutation-Operator Specialisations

B.1 Program Selection Rationale

(a) Library-stack coverage. numpy 2.4.4 (A1-B3 linear algebra / arrays), scipy 1.17.1 (A1 integrate, A2 linalg, B2 stats), scikit-learn 1.8.0 (C1-D3 surrogate / ML), Python scientific-computing's de facto foundation stack (PyPI top-50, 2026-04). Uncovered: GPU / distributed (JAX, CuPy, dask), domain-specific (BioPython, Astropy, RDKit). Left to future work under the representativeness threat.

(b) Mathematical-structure coverage. ODE/PDE (A1, A3), direct linear algebra (A2), Bayesian analytic + MCMC + Monte Carlo (B1-B3), kernel + orthogonal polynomial + NN surrogate (C1-C3), convex optimisation + backprop + max-likelihood (D1-D3), covers 8 of 12 chapters in Numerical Recipes [6]. Uncovered: advanced PDE solvers (FEM, FV, spectral); FFT; interior-point/trust-region optimisation; symbolic/CAS. Left to future work.

(c) Comparison with benchmarks. DeepCrime (Humbatova et al. 2021): semantic class-D topical overlap on ML kernels. Defects4J (Just et al. 2014): no overlap, only mutation-as-fault-proxy reference. mutmut / cosmic-ray demos: general Python; the semantic operators extend the scientific-computing focus.

(d) Scale vs representativeness. Each PUT 50-400 LOC; signature standardised to `program(x: float) → float`. Substantively smaller than industrial code (typically 1-10 KLOC).

Limitation. The `float → float` signature is a substantive constraint, not engineering trade-off. Industrial inputs are typically high-dimensional (CFD grids, MD particles, FEM matrices); scalarised PUTs upper-bound mutant semantic complexity and may systematically under-estimate SMS and cross-class differences on industrial PUTs (declared in the main manuscript's threats-to-validity section; future work validates on industrial-grade PUTs).

B.2 Systematic vs incidental

Syntactic tools may *occasionally* hit the semantic-mutation criteria in the main manuscript (a) (cross-function-boundary) or (c) (algorithm class), but occasionality undermines two engineering functions of semantic mutation.

(a) Deepening source-code understanding. Designing OS `np.linalg.det(M)` → `np.sum(np.diag(M))` requires knowing the two are equivalent on diagonal matrices but not on general matrices (`det` = $\prod$ `eigvals` vs `sum(diag)` = `trace` = $\sum$ `eigvals`); syntactic tool AST traversal does not require this understanding even when similar replacements appear. **(b) Revealing deep faults.** Domain-semantic errors (physical-constant, unit conversion, boundary conditions, hyperparameter semantics, numerical-method order) are not AST-local; syntactic mutator design goals (operator typos, off-by-one, negation flips) have hit probability for domain errors much smaller than systematic semantic-mutator triggering, and lack repeatability. **Conclusion.** Syntactic tools occasionally producing mutants satisfying (a)(b)(c) is stochastic byproduct, not repeatable, not carrying engineering value. Systematic semantic mutation requires (a)(b)(c) to be design intent.

B.3 Per-class operator specialisations

Table 3. Per-class operator specialisations.

Operator	Representative specialisations across PUTs
mut_C Conservation-breaking	A1 Lorenz: add $\varepsilon_{\text{drift}}$ to RHS (slow Hamiltonian drift); A2 LU: decomposition omits k+1-th multiplier (breaks det conservation); B1 Beta-Bin: posterior omits normalisation; C1 GPR: covariance omits positive-definite diagonal term; D1 MLP: backprop omits one gradient term.
mut_M Monotonicity-breaking	A3 FDM: Δt coefficient occasionally negative; B2 MCMC: acceptance min(1, r) → min(0.95, r); C2 PCE: high-order coefficient sort inserts inversion; D2 SVM: decision-function sign flips near boundary.
mut_G Convergence-breaking	A1 Lorenz: RK4 → 1.5-order hybrid; A3 FDM: 2nd-order difference → 1st-order; B3 MC: doubling sample size does not 1/N-reduce variance; C3 NN-Surr: training-epoch truncation.
mut_T Trajectory-distorting	A1 Lorenz: state-vector y/z swap; B2 MCMC: insert independent-sampling segment; C3 NN-Surr: training-target slow phase shift; D1 MLP: hidden-layer periodic-mask activation.
mut_F Fidelity-order-breaking	A2 LU: partial pivoting degrades to no pivoting; C1 GPR: length-scale switches to coarse prior; C2 PCE: high-order term randomly retains low-order; D3 LR: regularisation occasionally large.

Per-class HP / OS / TF substitution rules (illustrative). **HP** on class a: tolerance / max_iter; on class c: GPR `noise_level` / `length_scale`; on class d: MLP `hidden_dim` / dropout. **OS** on class a: numerical-linalg API swaps (`det`

↔ `sum(diag)`); on class b: probability-distribution sampling swaps; on class c: surrogate-class swaps (GPR ↔ RBF ↔ NN). **TF** on class a: integration-order changes (RK4 → Euler); on class b: MC estimator changes.

Operator-level cross-table with cosmic-ray defaults. The 12 cosmic-ray default operators are AST-local: `NumberReplacer` (Num/Constant, CE partial, △ literal values only); `ReplaceArithmeticOperator` (BinOp); `ReplaceComparisonOperator` (Compare); `ReplaceLogicalOperator` (BoolOp); `ReplaceUnaryOperator` (UnaryOp); `ReplaceTrueFalse` (NameConstant, CE keyword, △ Bool literal); `BreakContinueReplacer` (Break/Continue); `RemoveDecorator`; `RemoveExceptHandler`; `ZeroIterationForLoop` (For); `ReplaceIfBlock` (If); `MutateSubscript`. None correspond to OS / HP / TF / SI:

Table 4. Per-class operator specialisations.

Semantic class	Tool support	Coverage
OS API replacement	None	△ 88.33% disjoint (incidental low-complexity hits)
HP	None	× tool inexpressible (0/72)
TF numerical-method order	None	× tool inexpressible (0/54)
SI / **CF** structural injection	None	× tool inexpressible (0/33)

Operator-level conclusion. All 13 default classes remain AST-local (BinOp / Compare / BoolOp / UnaryOp / NameConstant / Number / Subscript / If / For / Break / Continue / decorator / except). Categorically, no entry recognises sklearn / scipy hyperparameter semantics (HP), numerical-method order (TF), or control-flow intent (SI / CF). For OS, low-complexity sub-expressions can be incidentally hit by BinOp; empirical 11.67% (the main manuscript's AST-overlap audit) is consistent with 88.33% AST-disjointness lower bound.

B.4 Syntactic-Mutation Pilot Study

Before the 12-PUT generalisation in the main manuscript's AST-overlap audit, a single-PUT lightweight empirical used `scripts/run_cosmic_ray_a1.sh` on a1 (file 934 B, AST-bound). Outcome: mutants_generated ~tens; ≥ 90% belong to BinOp / Compare / 13 default classes; small killed-by-`test_a1.py` ratio (PUT unit tests are output-shape/type sanity checks; most cosmic-ray mutants not killed), an empirical manifestation of the semantic-mutation criteria in the main manuscript: syntactic-tool mutants are not aligned with MR-violation detection. The 12-PUT generalisation (the main manuscript's AST-overlap audit) supersedes this pilot but corroborates the same conclusion at scale.

B.5 Expected Root-Cause Risk Profile

PUT-class × LRCA-layer risk weights. A numeric: C2 dominant (⋆⋆⋆ on L1 tolerance). B probabilistic: C4 dominant (⋆⋆⋆ on L3). C surrogate: C3 dominant (⋆⋆⋆ on L2 OOD; ⋆⋆ tolerance). D ML: C3 + C4 mixed (⋆⋆⋆ L2; ⋆⋆ L3).

Operator-PUT root-cause hotspots (expected). A: mut_C/G/F dominantly C2 (numerical tolerance), mut_M/T C1 (true semantic). B: dominantly C4 (statistical-assumption violations). C: mut_C/M dominantly C2/C1 on c1/c2, mut_T/F dominantly C3 on c1/c2; c3 (NN-Surr) row dominantly C5 (artefact). D: d1 row dominantly C5 (training-noise artefact); d2/d3 dominantly C1 / C3 mixed.

Expected suspect_share thresholds. A: 0.10-0.20 (acceptance ≤ 0.25); B: 0.20-0.35 (≤ 0.40); C: 0.20-0.30 (≤ 0.35); D: 0.25-0.40 (≤ 0.45). 60-cell average: ≤ 0.20 (= H4; acceptance ≤ 0.25).

B.6 Engineering-significance mapping

j = k aligned diagonal is the H2 threshold-test slice; j ≠ k off-diagonal is control; 9 vacant cells (∘, no MR exercised) are not formally adjudicated and are repurposed in the main manuscript's R_sem/R_kill decoupling discussion as descriptive evidence for R_sem / R_kill decoupling. The v3 → v4 micro-change ($\Delta\delta = -0.009$, paired-role bootstrap 95% CI [-0.238, 0.207]) attributes to LLM-source diversity under a fixed prompt; this is the source-axis null contrast underpinning the main manuscript's aligned-versus-cross results.

B.7 Default-Operator Overlap between Syntactic Mutation Tools

Reviewer-requested manual operator-class cross-reference between cosmic-ray's 13 default operators (cosmic-ray 8.4.6, `cosmic_ray.operators` package) and mutmut's 14 default operators (mutmut current `main`, `src/mutmut/mutation/mutators.py`). Both default-operator sets are first-order AST-local replacements; neither implements a cross-function-boundary, domain-knowledge-dependent, or algorithmic-class-changing mutation (the three necessary conditions in the semantic-mutation criteria of the main manuscript for semantic mutation).

Table 5. Mutmut vs cosmic-ray default-operator overlap.

Operator class	cosmic-ray default	mutmut default	Reaches the semantic-mutation criteria (a/b/c)?	Semantic-class reachability
Numeric literal ±1	`number_replacer`	`operator_number`	(a) no, (b) no, (c) no	partial CE only (e.g. `_RHO=28.0 → 27.5`)
Binary arithmetic swap (+, −, ×, ÷)	`binary_operator_replacement`	`operator_swap_op` (binary subset)	(a) no, (b) no, (c) no	partial OS only (incidental)
Comparison swap (<, ≤, >, ≥, ==, !=)	`comparison_operator_replacement`	`operator_swap_op` (compare subset)	(a) no, (b) no, (c) no	none
Boolean swap (and ↔ or)	`boolean_replacer`	`operator_swap_op` (bool subset)	(a) no, (b) no, (c) no	none
Unary remove / swap	`unary_operator_replacement`	`operator_remove_unary_ops`, `operator_swap_op`	(a) no, (b) no, (c) no	none
Keyword swap (True/False, etc.)	`keyword_replacer`	`operator_keywords`, `operator_name`	(a) no, (b) no, (c) no	none
break ↔ continue / return	`break_continue`	`operator_keywords` (subset)	(a) no, (b) no, (c) no	none
Exception class swap	`exception_replacer`	(none)	(a) no, (b) no, (c) no	none
Decorator removal	`remove_decorator`	(none)	(a) no, (b) no, (c) no	none
Variable-binding insert / replace	`variable_inserter`, `variable_replacer`	`operator_arg_removal`, `operator_assignment`	(a) no, (b) no, (c) no	none
no_op insertion	`no_op`	(none)	(a) no, (b) no, (c) no	none
Zero-iteration for loop	`zero_iteration_for_loop`	(none)	(a) no, (b) no, (c) no	none
String content tweak (XX prefix, case flip)	(none)	`operator_string`	(a) no, (b) no, (c) no	none
Lambda body → `None`/`0`	(none)	`operator_lambda`	(a) no, (b) no, (c) no	none
Dict kwarg name (XX prefix)	(none)	`operator_dict_arguments`	(a) no, (b) no, (c) no	none
Sym./asym. string-method swap	(none)	string-method symmetric swap; string-method asymmetric swap	(a) no, (b) no, (c) no	none
Augmented → simple assignment	(none)	`operator_augmented_assignment`	(a) no, (b) no, (c) no	none
`match` case removal	(none)	`operator_match`	(a) no, (b) no, (c) no	none

Aggregate. Cosmic-ray ∩ mutmut: ~9 of 13 cosmic-ray operators have a near-equivalent in mutmut (numeric, binary, compare, boolean, unary, keyword, break-continue, variable, plus partial overlap on string-prefix vs `XX`); 4 cosmic-ray operators are unique (exception, decorator, no_op, zero-iteration-for); 6 mutmut operators are unique (string content, lambda, dict-kwarg, string-method swap, aug-assignment, match-case). The union is approximately 21 distinct first-order AST-local operator classes. None of the 21 can:

- (1) Cross a function-call or module-import boundary (e.g. replace `det(M)` with `sum(np.diag(M))` requires knowing the equivalence on diagonal matrices, outside any default-operator scope);
- (2) Depend on domain knowledge for legality (e.g. a hyperparameter swap that knows `noise_level=1e-4` is the load-bearing knob in a Gaussian Process Regression PUT requires PUT-class awareness);

- (3) Change the algorithmic class (e.g. swap `polynomial` → `spline basis` rewrites the surrogate's mathematical structure).

The mutmut / cosmic-ray null intersection on the main manuscript's AST-overlap audit's HP/SI/TF classes (zero overlap) therefore reflects a **structural property of first-order AST-local mutation** rather than a tool-specific limitation, supporting the preventive-defence framing in the main manuscript.

C Experimental Procedure Details

C.1 Cross-Source Generation Protocol

Two-stage ablation.

Table 6. Cross-source generation protocol.

Version	Mutant pool	c-class primary MR	Use
v3	Single-source (Claude Opus 4.6)	P-ord. (pre-registered)	H2 baseline
v4	**Cross-source** (Claude Opus 4.6 + GPT-5.4 + DeepSeek chat)	P-ord. (held at pre-registered)	Isolate LLM-source diversity

Protocol (`scripts/cross_source_campaign.py`). (a) For each (PUT, operator), Claude / GPT / DeepSeek run K = 3 trials with identical prompt (the cross-source protocol in the main manuscript), temperature 0.7; `source_tag` propagated to filenames. (b) **V1-V4 mechanical validation** (`src/p2/mutators/validation.py`): V1 syntax (`ast.parse`), V2 executability, V3 non-triviality (`|y_mutant - y_original| > 1e-6`), V4 signature consistency. (c) **DeepSeek model:** `deepseek-chat` (113 tokens/call, 1.8 s), not `deepseek-v4-pro` (340 tokens/call, 11 s), quality-equivalent on a2_OS1 dry-run. (d) **Pool capacity:** 37 ops × 3 sources × 3 trials = 333 attempts; V1-V4 pass rate 89.5%; 298 confirmed mutants, of which 292 enter the v4 pool after final-stage rejection of duplicates and QA failures. Sources contribute nearly equally: Claude 101 / GPT 98 / DeepSeek 99 (Phase A engineering finding). (e) **Sampling** (`scripts/build_pools.py`, POOL_VERSION = v4): per-PUT 30 max, measured mean 24.3, range 10-30. c1 (GPR) has only 10 because c1_HP1 / c1_CE1 have V1-V4 near-zero pass (WhiteKernel noise_level 1e-4 → 1e-1 perturbation has minimal output impact, triggers V3 non-trivial failure, which supports the main manuscript's R_sem/R_kill decoupling discussion).

Protocol asymmetry. v4 does not invoke reviewer LLM (cost / speed priority; a follow-up study will rerun dual-blind on the v4 grid, ~$5-8 USD); v3 used the original Phase-1 dual-blind (Claude gen + GPT-5.4 review + DeepSeek arbitration). A fraction of the v3 → v4 source-axis shift may be slight v4 quality decline rather than LLM-source diversity.

C.2 Differential prompt protocol

The v3 → v4 micro-change of -0.009 (paired-role bootstrap 95% CI [-0.238, 0.207]) may reflect "three LLMs under identical prompt" rather than "true upper bound of LLM diversity". Separation requires *one tailored differential prompt per LLM* (skeleton: `scripts/run_differential_prompt.py`).

Design (2×3 factorial, within-(PUT, operator)). Factor A: prompt template, 3 levels: **V_canonical** (control, identical to v4), **V_persona** (per-LLM identity: Claude "numerical-analysis editor"; GPT "scientific-software refactorer"; DeepSeek "library-API substitution specialist"), **V_cot** (Claude <thinking> tags; GPT "step by step"; DeepSeek stepped-reasoning). Factor B, LLM source (Claude / GPT-5.4 / DeepSeek chat). K = 3 per cell; total 37 × 3 × 3 × 3 = **999 trials**.

Exit criteria. If max(delta) − min(delta) < 0.05: the v3 → v4 −0.009 source-axis null is robust. If ≥ 0.05 with prompt variant pushing delta past 0.474: H2 revised from "not met" to "prompt-conditional met". If ≥ 0.05 but all delta < 0.474: prompt sensitivity exists but H2 unchanged.

Resources. API ~$18-30 USD; wall 3-4 h at concurrency 4; V1-V4 only, no reviewer LLM.

C.3 Manual Pilot and Initial Prescreen

The original cross-source protocol used 60% multi-LLM consensus (Claude Opus 4.6 + GPT-5.4 + DeepSeek chat unanimous) plus 40% manual injection by scientific-computing researchers. Each (mut_j, S_i) pair carried one prompt template containing PUT source, semantic intent of mut_j, prohibition against revealing specific MRs (avoids prompt leakage), and output requirements (syntactically correct, executable, single-point modification, diff < 10 lines). Reproducibility parameters: temperature 0.3 in manual-pilot stage (v4 cross-source uses 0.7 per §C.1); fixed seed; 5 candidates per prompt with 2-3 retained.

Double-blind review (Scheme C). Generator LLM-G (Claude Opus) and reviewer LLM-R (GPT-5.4) are cross-source with strict role separation. LLM-R sees only PUT original + mutant code, unaware of mut_j category / MR content / generator identity, outputting a triple (syntactic, executable, semantic-fault-injected). Double-confirmed mutants enter pool; inconsistencies enter manual arbitration (≤ 10%). From the double-confirmed pool, 20% is sampled for manual review; manual-vs-LLM-R inconsistency > 10% triggers full manual downgrade. Same-source bias mitigation stratifies LLM-R by PUT class. Prompt-injection mitigation disables RAG / web search on the API path.

L0 pool prescreen. Each mutant passes static syntax (linters / type checkers), a unit self-test (simple inputs produce finite output), and double-blind sign-off. Target: 30-50 candidates per cell → 10-15 retained after prescreen.

C.4 Three-Layer Root-Cause Diagnosis

Decision tree, multi-label priority (C5 > C4 > C3 > C2 > C1), and output (`C1_share`, `suspect_share`) are stated in §A.2; the full pseudocode is reproduced there. LRCA does **not** modify SMS; killed set is not filtered. The main manuscript's LRCA-mass results use the best calibrated combination (`ood_band = 0.02, tolerance_multiplier = 3.0, repeats = 20`); default-threshold results retained in `lrca_60cell_v3.json` as control. Interface with the preventive-defence risk profile in the main manuscript: L0 prescan ↔ §C.2 dual-LLM double-blind; L1 tolerance ↔ N = 20 repetition subprocess; L2 OOD ↔ input-distribution definition; L3 statistical baseline ↔ pre-check protocol.

C.5 Pilot calibration

A 2026 Q3 operator-level pilot ran 12 PUTs × 37 operators (36 primary PUT-class operator pairs plus one CF structural-injection specialisation; the conceptual semantic family count remains five; 12 `is_key=True` at K=20, 25 at K=10; 470 trials total). Pilot precursor quantities: R_sem (V1-V4 ∧ operator_match pass rate), D_impl (median pairwise AST + literal + identifier Jaccard distance among confirmed mutants), R_kill (proportion of confirmed mutants killed by AVP under that PUT's primary MR). Aggregate (N=37): R_sem median 0.50 mean 0.468 (0/37 with R_sem=0); D_impl median 0.42 mean 0.392 (1/37 with D_impl ≈ 0); R_kill median 0.00 mean 0.189. All 37 operators produced ≥ 1 V1-V4 passing mutant, a necessary precondition for H1; the pre-registered H1 threshold itself ($\geq$ 5 non-equivalent mutants on $\geq$ 9/12 PUTs for $\geq$ 4 of 5 operator families) is nonetheless not met, as reported in the main text.

Key R_sem / R_kill decoupling pattern. R_sem $\geq$ 0.9 $\wedge$ R_kill = 0 combinations all on HP / TF / OS operators in c / d classes (e.g., c1_CE1 noise 1e-4→1e-1 with R_sem 1.00, R_kill 0.00; c3_HP1 relu→tanh, c3_TF1 max_iter 1000→5, d1_HP1 MLP α, d3_HP1 LR C). R_sem-high $\wedge$ R_kill = 1 combinations all on CE / OS operators in a / b classes (a2_CE1 LU det, a2_OS1 prod→sum, b1_OS1 α/β swap, d1_TF1 label flip). This is empirical evidence for R_sem / R_kill decoupling at cell level.

D Statistical Analysis Details

D.1 Semantic-Mutation Score Variants and Construction Lemmas

SMS_unfiltered (reviewer comparison metric):

$$\mathrm{SMS}_{i,k,j}^{\mathrm{unfiltered}} := \frac{|\mathrm{killed}_{i,k,j}|}{|\mathrm{mut}_j(S_i)| - |\mathrm{equiv}_{i,k,j}|}$$

(Same form as the primary metric; here `killed` does not distinguish C1 from C2–C5.)

The appendix of the reproducibility package provides cell-by-cell difference between SMS_unfiltered and SMS; if relative difference < 5%, this confirms LRCA does not affect robustness of primary conclusions.

Primary-table construction.

Table 7. SMS variants and construction lemmas.

RQ	Primary statistics	Reporting format
RQ3	inst_rate, equiv_rate, C1_share, survive_rate	60-cell heatmap + 4-class marginals
RQ4	aligned-SMS vs cross-SMS; sparse ◦ vs dense ••equiv_rate	Cliff's delta + odds ratio + 95% bootstrap CI
RQ5	ΔSMS_c (c $\in$ {A,B,C,D}); CV(ΔSMS)	sign test (df=3) + descriptive forest plot
RQ5 (desc.)	Spearman ρ + Kendall τ for SMS vs PC	scatter + dual-metric ranking comparison

Multiple comparisons. No cell-level family of p-values is claimed across the 60 cells (main text, statistical pipeline); the per-class Friedman tests carry a Bonferroni × 4 correction. N = 20 bootstrap intervals: 1,000-iteration 95% CI (B = 10,000 for the headline H2 delta CI).

D.2 Effect-Size Sensitivity and Transformation Invariance

H4 cutoff sensitivity. Dense grid scan (cutoff $\in$ {0.05, 0.10, ..., 0.50}, step 0.05) on v4 data (`scripts/h5_sensitivity.py`, `data/results/h5_sensitivity_v4.json`):

Table 8. Effect-size sensitivity and logit-transformation invariance.

cutoff	h5_cells_pass	h5_pass_ratio
0.05	12 / 60	20.0%
0.10	12 / 60	20.0%
0.15	12 / 60	20.0%
0.20 (paper)	**12 / 60**	**20.0%**
0.25	12 / 60	20.0%
0.30	12 / 60	20.0%
0.35	12 / 60	20.0%
0.40	12 / 60	20.0%
0.45	13 / 60	21.7%
0.50	13 / 60	21.7%

Conclusion. H4 verdict (not met) is intrinsic data property, independent of cutoff choice. v4 suspect_share distribution is severely bimodal: median 1.0, mean 0.79; 48 cross cells at suspect_share ≈ 1, 12 aligned cells in low end; the [0.20, 0.80] interior is nearly empty. Specific value 0.20 is **not load-bearing**.

Logit-transformation invariance. Cliff's delta is rank-based (function of $U/(n_1 \cdot n_2)$), mathematically invariant under any strictly monotone transformation (Romano 2006). Logit is strictly monotone on (0, 1), so $\delta_{\text{logit}} \equiv \delta_{\text{raw}}$ is a construction result. **Not additional robustness evidence**; H2 robustness threats come from the main manuscript's 60-cell distribution results zero-mass dominance, not metric-scale choice.

Vacant-cell sensitivity. Excluding the 9 vacant cells (cells with no exercised MR) leaves the aligned-versus-cross verdict unchanged. In v4 the analysis becomes n = 11 aligned and n = 40 cross with $\delta = 0.323$ (95% CI [-0.011, 0.648]); the matching v3 slice gives $\delta = 0.270$ (95% CI [-0.043, 0.595]). The exclusion widens the interval because the design is smaller, but it does not move H2 across the pre-registered Romano medium threshold.

Primary-MR withdrawal permutation-null. The auxiliary c-class check pools 15 SMS values, shuffles labels over the 3 × 5 (PUT, MR) grid, and applies the max-over-five withdrawal rule with 10,000 permutations (seed 42). The observed c-class mean is 0.314 versus null mean 0.349, one-sided $p = 0.989$ (`data/results/c_class_permutation_v4.json`). This supports treating the primary-MR convention as a protocol choice rather than as evidence of a hidden c-class inflation effect.

D.3 Power Analysis under Observed and Stipulated Effects

Plug-in bootstrap. With-replacement sample from observed aligned (n=12) and cross (n=48) v4 SMS pools; N_sim = 5,000, seed = 42 (`scripts/compute_rq2_power.py`, `rq2_power_v4.json`):

Table 9. Power analysis under observed and stipulated effects.

Threshold	Interpretation	Power
delta > 0.000	Any-effect	**0.997**
delta > 0.147	Small	0.966

delta > 0.330	Medium	0.759
delta > 0.474	**Large (H2)**	**0.423**

Sample-size sweep ($n_{\text{aligned}} \in \{6, 12, \dots, 60\}$, $n_{\text{cross}} = 4 \times n_{\text{aligned}}$): power for delta > 0 reaches 0.974 at n=6, 0.996 at 12, plateaus thereafter. RQ4 primary analysis has very low sample-size requirement; the bottleneck is the effect-size boundary itself, not sampling noise. Even at $\delta_{\text{truth}} \approx 0.474$, sample has only ~42% chance of detecting under plug-in; but this does **not** mean "insufficient power is the cause of H2 not being met", observed $\delta_{v3} = 0.323 < 0.474$, observed $\delta_{v4} = 0.314 < 0.474$; increasing sample only narrows CI.

Stipulated-alternative. Implemented in mixture-weight (`scripts/compute_rq2_power_stipulated.py`, N_sim = 2,000): SMS has heavy ties at zero (45/60 = 0), raw shifting is discontinuous (any $\varepsilon > 0$ jumps delta from 0.314 to 0.74). Mixture: aligned' = (probability w) sample from (observed_aligned + 0.001) + (probability 1−w) sample from observed_aligned, calibrate w so E[delta] ≈ 0.474. Calibrated w = 0.094, realised E[delta] = 0.4746.

Table 10. Power analysis under observed and stipulated effects.

Stipulated truth	Criterion	Power
0.474	$\hat{\delta} \geq 0.474$ (point-estimate)	**0.491**
0.474	95% CI lower > 0 (any-effect)	0.868

Even at $\delta_{\text{truth}} = 0.474$, the (12, 48) design returns $\hat{\delta} \geq 0.474$ in only ~49% of replications. This *supports* the main manuscript's aligned-versus-cross framing that "not met" is point-estimate, not effect-size.

D.4 Per-Class Friedman Tests

Per-class Friedman (3 SUTs × 5 MRs) with Bonferroni × 4: a chi^2=4.00 raw 0.406 adj 1.000, W=0.333; b chi^2=10.34 raw **0.035** adj **0.140**, W=**0.862**; c chi^2=5.60 raw 0.231 adj 0.924, W=0.467; d chi^2=5.00 raw 0.287 adj 1.000, W=0.417. After correction, no per-class result remains significant at family-wise alpha = 0.05. Even b-class W = 0.862 (Cohen 1988 large concordance) is insufficient at N = 3 per class. **H3 verdict rests on the main manuscript's cross-class consistency sign test**; per-class Friedman is sensitivity / descriptive only. Friedman main effect on full 60 cells (chi^2 = 16.76, p = 0.0022) confirms MR rank differences but is logically independent from H3 cross-class consistency.

Mixed-effects unavailability. Primary `sms ~ C(class) + C(operator) + C(class):C(operator) + (1 | put)` returned Singular (insufficient column rank for class × operator interaction; N=60 too small for 11-d fixed-effects + 12 PUT random intercepts). Fallback `sms ~ C(class) + C(operator) + (1 | put)` had PUT random-intercept variance hit boundary 0 (degenerates to OLS). Fallback class p-values (descriptive only): b vs a 0.275 / c vs a 0.892 / d vs a 0.991. RQ5 conclusion shifts to four-piece presentation (class means + sign test + Friedman + forest plot).

D.5 Pattern-coverage operationalisation

Per SUT, (MR_k, R_outcome ∈ {True, False}) binary-tuple coverage: 5 MRs × 2 outcomes = 10 cells, PC = #triggered / 10. Simplest baseline (per-SUT granularity, not distinguishing mutants). PC range over 12 SUTs: [0.500, 1.000], mean 0.733 (`paper_numbers_v4.json`). Pairing per PUT: Spearman rho = 0.163; Kendall tau = 0.136 (descriptive only; this row carries no inferential licence in the main manuscript's inference-permissions table, so no p-value is attached).

Power caveat. At n = 12, Spearman 95% CI ≈ [-0.5, +0.6] (Fisher-z approximation); cannot distinguish "zero correlation" from "moderate positive" or "moderate negative". The descriptive rho / tau mean "no correlation detected at n = 12", not "correlation does not exist".

Within-class descriptive. b2 PC = 1.0 with mean SMS = 0.067; b1 PC = 0.7 with mean SMS = 0.20. c3 PC = 1.0 with mean SMS = 0.14; c1 / c2 PC = 0.7-0.8 with mean SMS = 0.0. Within the same class, higher-PC PUT has lower SMS, contradicts naive "more PC kills more mutants" assumption. This pattern is consistent with orthogonality, but n = 12 cannot confirm it. A confirmatory study would need $n \geq 30$ and a PC definition that incorporates the mutant dimension.

E Stakeholder Deployment Considerations

E.1 Air-Gap Deployment Constraints

The stakeholder-analysis workflow in the main manuscript depends on external LLM API calls (Claude / GPT / DeepSeek), incompatible with most regulated air-gapped V&V workflows:

Table 11. Air-gap deployment constraints.

Domain	Standard	Air-gap requirement
Nuclear	IEC 60880	Air-gapped build for safety-critical software
Aerospace	DO-178C	Tool-qualified offline build for DAL-A/B; LLM API outside qualified-tool envelope
Medical	IEC 62304	Class C software lifecycle requires offline reproducible build
Automotive	ISO 26262	ASIL-D requires offline tool chain; cloud LLM non-compliant

Mitigation paths. (a) Self-hosted open-weight LLMs (Llama / DeepSeek local inference), capability gap vs API-served frontier models; quantitative impact on SMS untested. (b) Offline-cached pre-generated mutant pools with commit-hash + raw-response signature locking, adopters reproduce a published pool offline (this paper's package supports this), but generating new pools requires external LLM access. SMS for air-gapped industrial deployment is a future research direction.

E.2 Quarterly batch audit workflow

The original per-PR GitHub Actions YAML was removed because its PR-CI threshold 0.10 contradicted the Appendix E.3 audit threshold 0.20 and propagated stakeholder-facing CI assumptions into adopters' pipelines.

Replacement: quarterly batch audit:

Table 12. Quarterly batch audit workflow.

Step	Description	Cost
1	Offline mutant-pool generation (per PUT 24-30 × 12 PUTs ≈ 300)	LLM API \$5-15; ~30 min
2	`sms_campaign.py --track 2`	4-core laptop ~10-20 min
3	`run_lrca.py`	< 1 min
4	MR-design team review of MRs with aligned-cell SMS significantly below historical median (0.275)	1-2 h meeting

Quarterly: API ~\$10 + automation 30 min + manual 1-2 h ≈ 0.5 person-day. Affordable. Per-PR gating not recommended (LLM latency 5-30 s + cost + non-determinism + air-gap incompatibility); adopters needing per-PR should use coverage / unit-test gates. SMS does not replace domain-expert MR physical-reasonableness judgement nor product-requirements review.

E.3 Verification-and-Validation Documentation

This appendix subsection makes **no normative claim** toward NRC, FDA, or ISO 26262 review teams. Within single-output kernels scope, no traceable mapping exists to current IEC 60880 / ISO 26262 / DO-178C / ASME V&V 20-2009 normative bodies.

V&V documentation for scientific computing software (per ASME V&V 20-2009 §3 and similar guides) requires quantifiable test-adequacy evidence; current documentation often relies on code coverage + MR lists + SME signatures. SMS may appear as **research-grade supplementary evidence** alongside coverage and MR lists: (a) aligned-cell SMS per critical PUT; (b) LRCA three-layer diagnosis (C1 readout / C2-C5 attribution); (c) 60-cell matrix visualisation. Reviewers can independently run `REPRODUCIBILITY.md`.

Original threshold recommendations (aligned-cell SMS ≥ 0.20 / 0.30 + C1_share ≤ 0.20) were removed: no normative backing in IEC / ISO / ASME; misreads as enforce-ready. **SMS reported as descriptive supplementary evidence**, with adequacy judgements left to V&V reviewers case-by-case.

Conceptually complementary to ASME V&V 20-2009 §3 code verification (numerical-solver correctness) vs SMS (MR-set fault-detection adequacy). Substantial scale gap to multi-module CFD codes; incorporation into V&V standards body would require large-scale empirics + multi-year dialogue with ASME V&V 20 / IEEE 1012 / IEC 60880 committees. **No advocacy that SMS enter any normative certification system in 2027.** SMS does not replace SME signatures, FMECA, PIRT, system-level V&V, or ASME V&V 20 §3 numerical-solver verification, it is one link in the evidence chain, not a single-point qualification.

F Threats to Validity: Detailed Mitigation

F.1 Internal threats

Hoeffding-style false-equivalence bound (referenced by main-text Limitation 1). For a mutant whose true per-sample disagreement probability against the original program under D_S is at least p_0 (at the E2 tolerance), the probability that all K_{eq} independent E2 samples agree, producing a false equivalence verdict, is at most $(1-p_0)^{K_{\text{eq}}} \leq$

$\exp(-K_{\mathrm{eq}}\, p_0)$. At $K_{\mathrm{eq}} = 1000$: $p_0 = 0.005$ gives $\leq e^{-5} \approx 0.7\%$; $p_0 = 0.01$ gives $\leq e^{-10} \approx 4.5 \times 10^{-5}$. The bound conditions on p_0 and says nothing about mutants whose disagreement is concentrated below the E2 tolerance; those are governed by the E1 arm and remain the residual risk the K_eq sweep (deferred) would quantify.

Table 13. Internal threats.

Threat area	Threat	Mitigation	Residual
LLM generation	LLM generation reproducibility	§C.2 reproducibility parameters in package; dual-LLM cross-source + 20% manual; reproducibility rests on archived raw responses and frozen mutant pools (no user-facing seed)	LLM training data may be updated post-submission
Equivalence approximation	Probabilistic equiv (K_eq=1000)	the main manuscript's equivalence-judgement section declares probabilistic approximation; Hoeffding-style false-equiv bound	K_eq sweep ∈ {500,1000,2000} deferred to future work
Shared infrastructure	Shared-infrastructure dependency	AVP version pinned to the upstream commit hash; package embeds AVP source; the arXiv infrastructure technical report is disclosed in the main text	None within scope
Root-cause diagnosis	LRCA multi-label boundary	§A.2 priority C5 > C4 > C3 > C2 > C1; multi-label co-occurrence table in package; root_cause is "likely root cause" not definitive	C2-C5 may multi-trigger on B/D classes
Pool size	Mutant-pool size	Pool expanded to mean 17.4: delta moved 0.321 → 0.323, CI narrowed; **effect size is intrinsic ceiling, not pool dilution**	Some PUTs < 30 (cache-bound)

LLM non-determinism	LLM non-determinism	Multi-turn de-dup; K=10/20 repetitions; raw prompt+response committed for direct reuse	Claude Opus subscription lacks seed control
Protocol asymmetry	**Protocol-implementation gap v3 vs v4**	v4 used 3 providers (Claude 101 / GPT 98 / DeepSeek 99) without dual-blind; v4 C1_share 0.209 > v3 0.164 weakly opposes "v4 quality decline"	Complete separation requires a follow-up dual-blind rerun on the v4 grid

The protocol-asymmetry threat (dual-blind vs V1-V4 only) applies to the v3 $\rightarrow$ v4 source-axis contrast ($\Delta\delta = -0.009$, paired-role bootstrap 95% CI [-0.238, 0.207]).

F.2 External / construct / conclusion threats

Table 14. External / construct / conclusion threats.

Threat area	Threat	Class	Mitigation
Representativeness	12-PUT representativeness	External	follows the shared site selection; open class principle (cls extensible); scaling to MD/QC/CFD
Statistical power	Cross-class statistical power	Conclusion	H3 framed exploratory; mixed-effects unavailable (Singular at N=60); shifted to class means + sign test + Friedman + forest plot
LLM homogeneity	LLM homogeneity bias	Construct	§C.2 three-LLM rotation + PUT-class subdivision + 20% manual sampling; third-party LLM family validation deferred

LLM-source shift	LLM-source distributional shift	External	Hit distribution DeepSeek 11/15, Claude 4/15, GPT 0/15; the main manuscript's AST-overlap audit argument is categorical AST-locality, not hit frequency; HP/SI/TF zero-overlap across all 3 sources unaffected
Higher-order mutation	HOM equivalence	External	Tool-unreachability claim confined to first-order syntactic tools; HOM empirical testing listed as future work

Conclusion threats: multiple comparisons handled per the main-text policy (per-class Bonferroni × 4; no cell-level family claimed; §D.1); N=20 stability handled by 1,000-iteration bootstrap CI (§D.3). Construct threat, SMS measures epistemological semantic detection, not production engineering value (out of scope here).

G Degeneration to Classical Mutation Score: Full Proof

G.1 Notation cross-reference

PUT S_i, mutation operator family mut (syntactic or semantic), mutant set $\mathrm{mut}(S_i)$, MR-label set, equivalence tolerance $\varepsilon_{\mathrm{eq}}$, equivalence sample size K_{eq}, AVP tolerance $\varepsilon_{\mathrm{AVP}}$. Three-state decomposition: $\mathrm{mut}(S) = \mathrm{killed} \cup \mathrm{equiv} \cup \mathrm{survive}$ (disjoint). SMS formula:

$$\mathrm{SMS}_{i,k,j} = \frac{|\mathrm{killed}_{i,k,j}|}{|\mathrm{mut}_j(S_i)| - |\mathrm{equiv}_{i,k,j}|}$$

G.2 Degenerate-limit definition

The degenerate limit $L = L_{\mathrm{equiv}} \wedge L_{\mathrm{killed}} \wedge L_{\mathrm{mut}}$ consists of three **joint conditions** (paired axes), each controlling one layer of the SMS formula. L1-L6 are not 6 independent axes; the pairing reflects dependency structure across the SMS formula layers.

Table 15. Degenerate-limit definition.

Joint condition	Component axes	Pairing rationale
L_{equiv} (Lemma G.1)	L1: $\varepsilon_{\text{eq}} \to 0$; L2: $K_{\text{eq}} \to \infty$	L1 alone: equiv stays probabilistic (K_{eq} cannot cover D_S). L2 alone: bitwise equality diluted by ε_{eq}. Both simultaneously degenerate equiv to classical behavioural equivalence outside a D_S-measure-zero set.
L_{killed} (Lemma G.2)	L3: $\varepsilon^k_{\text{AVP}} \to 0$; L4: MR-label set = $\{\text{MR}_{\text{eq}}\}$, R(y,y') ≡ y = y'	L3 alone: $\varepsilon_{\text{AVP}} \to 0$ still allows non-trivial MR (monotonicity, convergence). L4 alone: equality still carries ε_{AVP} tolerance. Both simultaneously degenerate killed to classical difference detection.
L_{mut} (Lemma G.3)	L5: $\text{mut}_j \to$ rule-based syntactic (Mothra AOR/ROR/SDL/CRP); L6: cls(I) ⊆ {imperative deterministic}	L5 alone: syntactic ops on probabilistic/ML may still trigger domain-semantic subsets. L6 alone: imperative programs still mutable by OS/HP/TF/SI. Both simultaneously degenerate mut(S) to Jia & Harman syntactic mutant set.

G.3 Lemmas for Three-State Decomposition

Lemma G.1 (equiv degeneration; measure-zero qualification). Assume **hypothesis NS** (D_S-non-singularity of the MR input transformations): no input transformation r used by $\text{MR}_{i,k}$ maps a D_S-positive-measure set into a D_S-measure-zero set; equivalently, the preimage $r^{-1}(N)$ of every D_S-measure-zero set N is itself D_S-measure-zero. Under L_{equiv} (L1 ∧ L2) and hypothesis NS, semantic-class equivalence (E1 ∧ E2) degenerates to classical behavioural equivalence **almost everywhere** w.r.t. measure D_S.

Proof. Both arms of the E1 ∧ E2 judgement carry the canonical definitions of §A.1 (E1: AVP coherence, $\forall mr \in \text{MR}_{i,k} : \text{AVP}(S_i, mr) = \text{AVP}(s', mr)$; E2: bounded output equivalence), and only L_{equiv} is assumed. *E2 arm.* The check $\|S_i(x) - s'(x)\| \leq \varepsilon_{\text{eq}}$ for K_{eq} samples under L1 ($\varepsilon_{\text{eq}} \to 0$) ∧ L2 ($K_{\text{eq}} \to \infty$ with measure-equivalent sampling) is almost-everywhere equivalent to $\forall x \in D_S \setminus N : S_i(x) = s'(x)$, where N is a D_S-measure-zero set (continuous D_S: floating-point pathological points / NaN propagation; discrete D_S: $N = \emptyset$). *E1 arm.* In the same limit E1 adds no further restriction almost everywhere: if $S_i = s'$ outside N, every AVP evaluation reads identical outputs on source inputs and on their MR-transformed images outside N (by hypothesis NS the MR transformations r are D_S-non-singular, so they do not map positive-measure sets into N), giving $\text{AVP}(S_i, mr) = \text{AVP}(s', mr)$ for all $mr \in \text{MR}_{i,k}$ up to the same measure-zero set; conversely, since E1 ∧ E2 ⊆ E2, the conjunction never admits a mutant the E2 arm excludes. Hence E1 ∧ E2 degenerates to the pointwise-equality relation almost everywhere, which matches Jia and Harman [4] §3 classical equivalent-mutant definition under measure-zero equivalence classes. No component of L_{killed} or L_{mut} is used.

Lemma G.2 (killed degeneration). Under L3 ∧ L4, killed degenerates to classical difference detection.

Proof. L4 restricts the MR-label set to {MR_eq} with R(y, y') ≡ y = y'. For mr = (r, R) and mutant s', the $\mathrm{MR}_{\mathrm{eq}}$ violation condition $\exists x : S_i(x) \neq s'(r(x))$ under L3 ($\varepsilon_{\mathrm{AVP}} \to 0$) becomes exact inequality. With r = id, violation is $S_i(x) \neq s'(x)$, classical difference detection. With r ≠ id, $\mathrm{MR}_{\mathrm{eq}}$ still requires $S_i(x) = s'(r(x))$ as reference oracle from original program; no new state classifications introduced.

Lemma G.3 (mut degeneration). Under L5 ∧ L6, mut_j(S_i) degenerates to the syntactic mutant set of Jia and Harman [4].

Proof. L5 switches mut_j to rule-based syntactic operators (AOR, ROR, SDL, CRP, UOI, standard Mothra/Proteum sets); L6 restricts PUTs to imperative deterministic programs, excluding triggering conditions for semantic operators on probabilistic/ML programs. Under this configuration, $\mathrm{mut}_j(S_i)$ is the literature-defined syntactic mutant set, independent of domain semantics.

G.4 Degeneration Theorem: Detailed Proof

Theorem G (SMS → MS degeneration; stated in the main text as the degeneration theorem). In the degenerate limit $L = L_{\mathrm{equiv}} \wedge L_{\mathrm{killed}} \wedge L_{\mathrm{mut}}$, and provided the MR input transformations are D_S-non-singular (hypothesis NS of Lemma G.1), almost everywhere w.r.t. D_S,

$$\mathrm{SMS}_{i,k,j} \xrightarrow{L} \mathrm{MS}_{i,j} := \frac{|\mathrm{killed}_{i,j}^{\mathrm{classic}}|}{|\mathrm{mut}_j^{\mathrm{syntax}}(S_i)| - |\mathrm{equiv}_{i,j}^{\mathrm{classic}}|}$$

Proof. Combine Lemmas G.1–G.3, each under exactly its own joint condition (Lemma G.1 additionally under hypothesis NS):

- Numerator: under L3 ∧ L4, $\mathrm{killed}_{i,k,j} \to \mathrm{killed}_{i,j}^{\mathrm{classic}}$ (Lemma G.2); L4 makes $\mathrm{MR}_{i,k}$ trivial over k (only $\mathrm{MR}_{\mathrm{eq}}$ remains), so subscript k degenerates.
- Denominator $|\mathrm{mut}_j(S_i)|$: under L5 ∧ L6 → $|\mathrm{mut}_j^{\mathrm{syntax}}(S_i)|$ (Lemma G.3).
- Denominator $|\mathrm{equiv}_{i,k,j}|$: under L1 ∧ L2 → $|\mathrm{equiv}_{i,j}^{\mathrm{classic}}|$ (Lemma G.1 under hypothesis NS; almost-everywhere w.r.t. D_S).

Substituting into the SMS formula gives $\mathrm{MS}_{i,j}$.

Remark (LRCA layer). The diagnostic LRCA layer sits outside the SMS formula, so Theorem G carries the backward-compatibility claim on its own. Whether the root-cause inventory C = {C1, ..., C5} also collapses to {C1} under L depends on the engineering thresholds of the §A.2 decision tree; no formal claim is made.

Empirical consistency. Theorem G shows that the SMS-based empirical conclusions (Cliff's delta in the aligned-versus-cross results, Friedman chi^2 in the cross-class consistency results, and Spearman rho in the SMS-vs-Pattern-Coverage analysis) are structurally consistent with existing Jia and Harman [4] literature in classical syntactic-mutation scenarios, no metric-level semantic fragmentation.

H Adjoint Extension Study

This appendix gives the full specification of the confirmatory adjoint extension arm summarised in the main manuscript. The arm exercises the adjoint invariant ψ_6, the MR family $\mathrm{f}_{\mathrm{adj.self}}$ under the MetaPattern m_{adj} (self-adjoint operator T^*) in NOETHER's taxonomy [1], on three third-party linear-operator libraries, disjoint from the 12 PUTs and the pre-registered 60-cell. Each library carries a real fixed defect in its adjoint code path as a ground-truth anchor:

- **scipy.sparse.linalg LinearOperator** (linear algebra; BSD-3). Program under test: the scaled-operator adjoint M^{H} for $M = \alpha A$ with complex scale α. Real anchor: scipy issue #8900 / PR #8962 (the adjoint dropped the conjugate on α, returning αA^{H} instead of $\bar{\alpha} A^{\mathrm{H}}$).
- **pylops.signalprocessing.FFT** with `real=True` (signal processing; BSD-3). Program under test: the real-FFT adjoint `_rmatvec`; the adjoint identity is the library's own *dot-test*. Real anchor: pylops issue #268 / PR #292 (commit `bd3a919`), a wrong scaling of the middle/Nyquist frequency bin that made the adjoint inconsistent with the forward.
- **jax** `linear_transpose` / `vjp` (automatic differentiation; Apache-2.0). Program under test: the autodiff-derived transpose M^{T} of a linear operator containing an inverse real FFT, $M(x) = \mathrm{irfft}(\mathrm{rfft}(x) \cdot w)$. Real anchor: jax issue #6223 / PR #6232 (commit `112ae41`), an `irfft` transpose rule that reversed order with an off-by-one index, so `vjp` silently produced a non-adjoint transpose.

For each subject the MR batteries are adj.self (self-adjointness) and adj.dual (scaled/dual-operator transpose). The mutant pool spans adjoint-breaking fault mechanisms (drop-conjugate, sign flip, scale and phase errors, transpose-without-conjugation, Nyquist mis-scaling), with the real fixed defect included among them, plus semantically equivalent mutants (scalar-multiplication reorder, conjugation of a real adjoint output) that a strong MR must *not* kill. Because the pre-fix releases are uninstallable on the host platform (no `arm64` wheel), each real defect is reproduced in extracted-diff form on the current release, mirroring the boundary study; $\varepsilon_{\mathrm{tol}}$ is an explicit factor throughout.

Table 16. Adjoint extension study, forward kill.

Substrate	Operator class (real anchor)	Active killed / total	Equiv. survived	Forward SMS
scipy LinearOperator	linear algebra (#8900)	5 / 5	2 / 2	**1.00**
pylops FFT real	signal processing (#292)	4 / 4	2 / 2	**1.00**
jax vjp / transpose	automatic differentiation (#6223)	4 / 4	2 / 2	**1.00**

Across all three substrates the correct program survives, every non-equivalent (active) mutant is killed, including the three real fixed defects, and no semantically equivalent mutant is killed, so the forward SMS is 1.00 with zero false positives on the equivalent set. The scipy #8900 defect is shared with the strong-boundary study in the main manuscript (adj.self case) and is not counted as independent corroboration twice; the pylops #292 and jax #6223 defects are unique to this arm. The pools are hand-constructed and low-cardinality, so the arm demonstrates the duality forward rather than measuring a high-adequacy MR set on a discovered fault population (main-paper adjoint-arm reproduction-fidelity threat).

I Real-Defect Evidence Summary

This appendix records only the result-level facts used by the main manuscript's real-defect evidence slice. It deliberately excludes candidate mining, rejected cases, repository tooling, and benchmark-release narrative. The purpose is auditability for the semantic-mutation argument: kill-rate, semantic-stratum alignment, and real-defect detection are related but distinct constructs.

Table 17. Result-level real-defect evidence summary.

Audit item	Evidence content used in the manuscript	Main-text role
Verified stratum counts	34 verified MR-detectable defects across the five NOETHER MetaPatterns [1]: m_{mono} 10, m_{conv} 9, m_{inv} 6, m_{adj} 5, and m_{rev} 4. This real-defect corpus is classified at NOETHER's Layer-1 MetaPattern level, whereas the MP1–MP5 experimental grid is stratified at the finer Layer-2 family level ($\mathrm{f}_{\text{inv.con}}$, $\mathrm{f}_{\text{mono.stat}}$, $\mathrm{f}_{\text{conv.lim}}$, $\mathrm{f}_{\text{mono.shape}}$, $\mathrm{f}_{\text{conv.rate}}$) and does not touch m_{rev}, so these five MetaPatterns are not the MP1–MP5 strata.	Shows that the declared semantic strata are not only toy labels
Mutation-phase aggregate	The semantic-mutation track exceeds the closest baseline by +0.101 on mean score (Holm-adjusted $p = 0.046$, Cliff's $\delta = +0.247$); the remaining pairwise contrasts are not significant	Supports a modest, qualified mutation-score advantage rather than a dominance claim
Real-defect face	The semantic-mutation track detects all 34 tabled real defects; the two generic baseline groups have nonzero detection in 7/34 and 8/34 cases, and the two ablated variants miss 19/34 and 17/34 cases, respectively, with 11 shared misses	Separates real-defect detection from aggregate mutant kill-rate
Non-nesting cases	Four verified non-nesting defects (A-LAPACK-004, A-OPENBLAS-001, B-POCKETFFT-002, and E-ORDINARYDIFFEQ-001) spanning linear algebra, FFT, and ordinary-differential-equation solvers	Supports the semantic-effect fiber view rather than a single scalar hierarchy

This summary is intentionally a result table, not a corpus or benchmark description. The main manuscript uses it only to support construct separation: a relation battery can score similarly on aggregate mutant kill-rate while differing sharply on real-defect detection, and different relation batteries can observe non-nested semantic-effect fibers.

J Study 2 Confirmatory Audit: Per-Family, Per-Class, and Incident Detail

This appendix records the per-family and per-class detail behind the Study-2 confirmatory scoreboard in the main text, the calibration and confirmatory incident log, and the review-packet protocol. The registration of record is `PREREGISTRATION_STUDY2_v1.1.md` and the append-only pilot log is `PILOT_LOG.md`; both are archived with the reproducibility package. Every number below traces to a pre-frozen SSOT.

J.1 Operator instantiability (H1′)

For each operator family, Table 18 gives the count of the 28 confirmatory PUTs on which the family produced ≥ 5 non-equivalent admitted mutants, against the outcome-independent coverage ceiling from the operator registry. The registered bar is ≥ 4 of 5 families clearing ≥ 8 of 28 PUTs; all five clear. Source SSOT `h1_instantiability_v5.json`. Pilot PUTs $\{a2, b4\}$ are excluded.

Table 18. Study-2 H1′ per-family instantiability on the 28 confirmatory PUTs.

Family	Coverage ceiling	PUTs cleared	Clears ≥ 8?	Note
CE	23	23	yes	clears every applicable PUT
OS	14	14	yes	clears every applicable PUT
HP	21	21	yes	clears every applicable PUT
TF	15	15	yes	clears every applicable PUT
SI	10	8	yes	narrow family; registration expected a possible miss, clears at 8
Total	—	5/5	—	bar is 4/5; verdict CONFIRM

J.2 Cross-class direction consistency (H3′)

Table 19 gives the class-mean aligned and cross SMS and the resulting direction for each of the four design classes, with the descriptive (under-powered, non-confirmatory) per-class binomial sign-test p. Three of four classes are positive; class C (surrogate models) is negative and reported as such. Source SSOT `h3_class_consistency_v5.json`. The across-meta-pattern Friedman statistic is $\chi^2 = 26.23$ over 28 blocks and 5 treatments, exploratory only.

Table 19. Study-2 H3′ per-class direction.

Class	n PUTs	Mean aligned SMS	Mean cross SMS	Direction	Sign-test p (desc.)
A	7	0.3333	0.0119	positive	0.0312
B	6	0.2222	0.0903	positive	0.1875
C	7	0.0476	0.0714	negative	0.9688
D	8	0.3333	0.0937	positive	0.0352

3/4 classes positive; bar is $\geq 3/4$; verdict CONFIRM. Per-class p values are descriptive only.

J.3 Attribution purity (H4′)

Table 20 gives the per-family multi-stratum leakage underlying the H4′ miss. The mean `suspect_share` over the 140 confirmatory cells is 0.1714, above the registered 0.05 bar. CE and HP stay clean; the leakage concentrates in TF and OS, with CF and SI contributing the remainder. Source SSOT `s5_purity_v5.json`; LRCA machinery is imported unchanged from the Study-1 S5 audit.

Table 20. Study-2 H4′ per-family multi-stratum leakage.

Family	Mutants scored	Multi-stratum	Note
CE	207	0	clean
HP	189	0	clean
OS	123	27	newly leaks (0 in Study 1)
TF	135	72	leaks most; single-stratum spec not contained on widened grid
CF	18	9	residual leakage
SI	69	9	residual leakage
Mean `suspect_share` 0.1714 > 0.05 bar; verdict NOT_CONFIRMED; offending families CF/OS/SI/TF.			

J.4 Incident and deviation log

Table 21 summarises the calibration-pilot and confirmatory incidents. The pilot phase (P1–P5) fixed code defects only, under the registration firewall (no threshold, estimand, DGP, primary-meta-pattern, or roster change). The confirmatory phase carried two tooling incidents (P6, P7) with no protocol impact, one disclosed script-interface deviation (D-A1), and one disclosed design-intent defect (P8) that leaves the registered decision rule intact but was not realised as intended.

Table 21. Study-2 incident and deviation log.

#	Class	Summary
P1	Near-miss (recovered)	Pool-build tool ignored CLI arguments and wiped the frozen Study-1 v3 pools from an empty cache; recovered by `git restore`; no artefact lost, no commit reached.
P2–P3	Code defect (pilot)	Pool builder gained argument parsing, a v5 version map (N=30, cross-source cache), and four delete-safety guards (empty-cache refusal, no empty-overwrite, wrong-version and frozen-version deletion guards).
P4	Code defect (pilot)	Response filename convention not stated in packets; an explicit `response_filename` field added to generation/review/arbitration packets.
P5	Code defect (pilot)	$E1 \wedge E2$ equivalence semantics not defined in-packet; operational definitions embedded verbatim in the blinded review and arbitration instructions.
P6	Tooling incident	Scoring tool defaulted to the Study-1 track and overwrote `sms_track1.json`; restored before use and rerun with an explicit track-2 target. Deterministic re-measurement of frozen pools, not regeneration.
P7	Blinding guard fired	One PUT's mutants carried a source-family token in docstrings; presentation-layer redaction added at export (displayed code only), affected packets re-exported, assertion retained as the final guard.
D-A1	Disclosed deviation	Pre-frozen δ script gained a post-data mode that skips only the gated H2-2 computation and emits its registered not-run verdict; H2-1′ logic byte-unchanged; recorded in the output SSOT.

P8	Disclosed design-intent defect	The CF/TF single-stratum admission screen was wired but a silent no-op for the whole v5 campaign: the pool builder tagged op-ids with a model-source suffix (e.g. `c7_TF1_claude`) that the screen's category regex rejects, so category lookup returned `None` and every candidate was admitted without a flip-count check (81/81 CF/TF null-pass; the pilot's "9/9 pass" was the same false negative). H4′ thus evaluated the *un-screened* pool. The registered decision rule scores `suspect_share` on the admitted pool and does not condition on the screen, so the frozen NOT_CONFIRMED verdict is valid; only the design intent (CF/TF entering single-stratum) was not realised. The filter code is deliberately *not* fixed here; correcting it belongs to the Study-3 registration wave. See the post-hoc diagnosis in the main text.
P9	Code defect (Study-3 wiring)	The build/score tooling (`build_pools`, `sms_campaign`) knew only pool versions v2–v5, so a v6 request had no suffix/cache/N mapping and no strict pool resolution (the same omission class as v5's D2/D4, on the Study-3 side). The fix mirrors the D2/D4 wiring: it adds `v6` (`_pool_v6`, cross-source cache, N = 30, strict resolution) and builds v6 under the live all-family screen, with a wrong-version deletion guard so a v6 build can only ever remove a `_pool_v6` directory and the frozen v5 pools are never touched. This is the only post-freeze code change: it encodes the registered N for a new pool version and points scoring at the correct strict directory, changing no threshold, estimand, primary-meta-pattern assignment, or roster, and reading no Study-3 outcome.

J.5 Review-packet protocol

Confirmatory review used the same dual-blind pipeline as the registered protocol: generation, then blind review on packets that expose only the mutant code, the operator spec, and the PUT source (generator identity, arm label, and SMS withheld), then arbitration on disagreement, then a freeze-then-score step in which SMS is computed only after review labels are frozen. The harness spawns each reviewer and the arbiter as separate, role-isolated Claude instances, so generator, reviewer, and arbiter differ on every item; all instances share one model family (same-vendor). The confirmatory pools' per-mutant source tokens `claude`/`gpt`/`deepseek` are nominal slot labels inherited from the Study-1 slot layout, all served by the same Claude-family generator, not real GPT or DeepSeek outputs; the aligned-versus-cross split is meta-pattern-based and independent of these slot labels, so no confirmatory statistic partitions by the vendor-named slot. In the confirmatory run, 18 independent reviewer instances returned 747 verdicts, 4 mutants were rejected at the first round, and 6 uncertain cases were arbitrated by a third instance and all resolved as confirmed under one uniform rationale (pinning the Gaussian-process hyperparameters before comparison). Mechanical admission gates V1–V4 admitted 756 of 774 valid mutants, yielding 140 confirmatory cells (36 with nonzero killed mass). Because the harness is same-vendor, the cross-vendor dual-blind hypothesis H2-2 is gated not-run, with no same-vendor proxy substituted.

J.6 Study-3 graded-attribution follow-up

Study 3 re-registers the attribution construct alone under `PREREGISTRATION_STUDY3_v2.md` (v2.0); the append-only pilot log is extended in the Study-3 section of `PILOT_LOG.md`. It replaces the single Study-2 purity bar with two Holm(2) verdicts in Family G and re-opens no other Study-2 verdict. The confirmatory statistics trace to the pre-frozen

SSOTs `sms_track2_v6.json` (fresh validated pool) and `h4_graded_v6.json` (both verdicts); the power reference is `power_study3.json`. Pilot PUTs $\{a2, b4\}$ are excluded from every confirmatory statistic.

Table 22 records the one-shot execution, and Table 23 the two registered verdicts with their per-class and per-family breakdowns.

Table 22. Study-3 confirmatory execution (one shot, seed 20260708).

Stage	Value
Valid generation records	765
Admitted (gates V1–V4, live all-family screen)	720
Rejected at admission	45 (families a3, b3, c1, d8)
Removed at pool assembly (720 → 633)	87: 81 by the single-stratum screen (nine PUTs each shed one full multi-stratum operator family of nine, e.g. d8 18 → 9), 6 by the registered per-PUT cap of 30 (a1 only, 36 → 30, OS/SI −3 each)
v6 confirmatory pool	633 mutants over 28 PUTs
v6 pool including $\{a2, b4\}$ pilot	678 files over 30 directories
Confirmatory cells	140 (20 with nonzero killed mass)
Blind-review verdicts	720 (0 schema errors, 0 arbitrations)
Live-screen re-audit	633 candidates matched, 36 flagged multi-stratum (v5 no-op matched 0)

Table 23. Study-3 Family-G verdicts (Holm 2).

Hypothesis	Threshold	Verdict	Breakdown
H4″-strict	purity ≥ 0.90 (95% lower CP)	CONFIRM	purity 1.0, CP-lower 0.9673 over $n = 90$ detected clean-family mutants (CE 72, HP 18, CF 0); screen matched > 0, gate passes
H4″-graded	rich share ≥ 0.15 (95% lower boot)	NOT_CONFIRMED	rich-class mean share 0.0833, lower bound 0.0, $n_{\text{rich}} = 6$ (C3–C6, D2, D5); per-class C 0.0 (4 PUTs), D 0.25 (2 PUTs)

Live screen, P8 remediation verified. The P8 fix (op-id parser tolerant of the model-source suffix, all-family scope, loud-fail on a null category) is exercised in the confirmatory campaign, not only in regression tests. The all-family re-audit over the 633-mutant pool matched all 633 candidates and flagged 36 as multi-stratum (`n_screened_candidates` 633, `n_multistratum_flagged` 36 in `h4_graded_v6.json`); the byte-identical v5 screen matched zero, so the registered screen-smoke gate, which requires a positive match, passes loudly rather than admitting an unconstrained pass.

Defect P9 (only post-freeze code change). The pilot surfaced one code-level defect: `build_pools` and `sms_campaign` knew only pool versions v2–v5. The fix, mirroring the v5 D2/D4 wiring, adds a v6 version (suffix `_pool_v6`, cross-source cache, N = 30, strict pool resolution) and builds v6 under the live all-family screen; a wrong-version deletion guard means a v6 build can only ever remove a `_pool_v6` directory, so the frozen v5 pools are never touched. It encodes the registered N for a new pool version and points scoring at the correct strict directory; it changes no threshold, estimand, primary-meta-pattern assignment, or roster, and reads no Study-3 outcome.

K Study 1 Extended Protocol and Derivations

This appendix carries protocol prose, formal derivations, and long tabular material demoted from the main text during length consolidation. Nothing here is new: each item is referenced from the corresponding main-text location, and every number cited in the main narrative is retained in place. The appendix collects the full threats-to-validity table (K.1), the complete semantic-mutation theory derivations and theorem proofs (K.2), the full higher-order-mutation reachability analysis (K.3), and the Study-1 stipulated-alternative power detail (K.4).

K.1 Full Threats-to-Validity Table

The main text (the main threats-to-validity section) gives a five-row summary of the threat areas; the complete per-threat table, including the Study-2, Study-3, and Study-4 harness, screen-remediation, serving-stack, reviewer-reliability, and attribution disclosures, is reproduced here in full.

Table 24. Threats to Validity (full).

Threat area	Threat	Class	Mitigation summary	Detail
LLM generation	LLM generation reproducibility	Internal	Archived raw responses + frozen mutant pools (no user-facing seed; reproducibility by archival)	Appendix F.1
Equivalence approximation	E2 probabilistic approximation	Construct	K_eq = 1000 + Hoeffding bound; sweep deferred to future work	Appendix F.1
Shared infrastructure	Shared-infrastructure dependency	Internal	AVP version pinned to the upstream commit hash; embedded source	Appendix F.1
Root-cause diagnosis	LRCA multi-label boundary	Internal	Decision-tree priority + multi-label co-occurrence table	Appendix F.1
Representativeness	12-PUT representativeness	External	shared infrastructure; Appendix B.1 coverage argument; scaling to industrial PUTs	Appendix F.2

Statistical power	Cross-class statistical power	Conclusion	H3 framed exploratory; Friedman fallback; mixed-effects unavailable	Appendix F.2
LLM homogeneity	LLM homogeneity bias	Construct	3-LLM rotation + 20% manual sampling	Appendix F.2
LLM-source shift	LLM-source distributional shift	External	DeepSeek 11/15 of overlaps; argument is categorical, not frequency	Appendix F.2
Pool size	Mutant-pool size	Internal	v3 mean 17.4, v4 mean 24.3 per PUT; effect intrinsic, not pool-dilution	Appendix F.1
LLM non-determinism	LLM non-determinism	Internal	De-dup, K = 10/20 repeats, raw-response store	Appendix F.1
Higher-order mutation	HOM equivalence	External	Confined claim to first-order syntactic tools; HOM testing deferred	Appendix F.2
Protocol asymmetry	v3 vs v4 protocol asymmetry (dual-blind reviewer)	Internal	Quality not down (C1_share 0.164 → 0.209); follow-up to rerun dual-blind on v4	Appendix F.1

Adjoint reproduction	Adjoint-arm reproduction fidelity and by-construction pools	Internal	Real defects reproduced through extracted diffs (no `arm64` wheel); arm is confirmatory on hand-constructed, low-cardinality pools, not a high-adequacy discovery; scipy #8900 is shared with the the strong-boundary section boundary study	the adjoint-extension section
Real-defect evidence	Industrial real-defect arm boundary	External	Used at result level to confirm construct separation; reusable benchmark construction deferred outside this manuscript	the industrial-construct-separation section
Same-vendor mono-culture (Study 2)	Generator and reviewer instances share one LLM family	Construct	Role isolation + packet blinding enforced across instances, but diversity of judgment is single-vendor; disclosed, and H2-2 gated not-run rather than proxied	the Study-2 harness section
Harness-agent instantiation (Study 2)	Confirmatory generation/review run through the Claude Code Agent harness	Internal	Blinded packets, role-isolated instances, third-instance arbitration, one-shot rule; resolved model id recorded in the campaign SSOT	the Study-2 harness section

Attribution scoring/admission gap (Study 2)	H4′ leakage (mean 0.1714) exceeds the 0.05 bar on the widened grid	Construct	Reported as a genuine miss; single-stratum scoring filter and admission filter disagree cell-by-cell; closing it is an open problem, not a promised study	the Study-2 interpretation section
Same-family equivalence correlation (Study 2)	Generator and reviewer instances share a model family, so equivalence (E1 and E2) opinions may be correlated	Construct	Equivalence verdicts feed admission, but the deterministic AVP and equivalence machinery remains the authoritative SMS gate, which bounds the threat; the cross-vendor arm H2-2 is gated not-run	the Study-2 harness section
Screen remediation verified in production (Study 3)	The P8 admission-screen no-op could recur silently	Internal	The Study-3 all-family screen is wired live with a loud-fail smoke gate; on the v6 campaign it matched 633 candidates and flagged 36 multi-stratum where the v5 screen matched zero, so the P8 fix is verified in production, not only in unit tests	the Study-3 execution section

Rich-class attribution remains open (Study 3)	Graded declared-stratum attribution did not reach the confirmation threshold on rich (surrogate/ML) classes	Construct	Reported as a registered NOT_CONFIRMED (rich-class mean share 0.0833 $<$ 0.15 bar) with $n_{\text{rich}} = 6$; a richer multi-valued attribution model is an open construct question, not a promised study	the Study-3 interpretation section
Serving-stack change (Study 4)	Amendment v1.2 moved the Claude-family roles from the gateway to the session harness mid-campaign after a quota event	Internal	Amendment pre-outcome and arm-symmetric (one reviewer sees both arms' blinded packets), so it cannot bias $\Delta\delta$; non-Anthropic generation and the `gpt-5.5` arbiter stayed on the gateway; no threshold, estimand, decision rule, or seed changed	the Study-4 design and execution sections
One-shot integrity (Study 4)	Incident P15: after a quota top-up, a resume defect redrew already-drawn gateway slots	Internal	Redraws quarantined unread; first draws restored byte-identical from the committed checkpoint; resume logic fixed before any Study-4 outcome existed, so the first draw remains canonical	the Study-4 execution section

Reviewer reliability (Study 4)	Blind review labels are not perfectly reproducible across reviewers	Construct	Split verdicts and opposite same-content labels recorded as reviewer-reliability data, with the mechanical AVP/equivalence pipeline authoritative for SMS; a post-hoc cross-vendor shadow review sides with the stricter literal reading, frozen labels unchanged (detail in the paragraph following this table)	the Study-4 execution section
H-LANG power and a2 retention (Study 4)	The C grid landed at 7 of 12 PUTs and retains a2, a Python-side pilot PUT	Conclusion / Construct	Power recomputed honestly at 0.6865 and disclosed at registration, no threshold moved; a2 kept confirmatory per amendment v1.1 A2 (C-side data fresh; {a3, b2} used for the C pilot); the registered a2-excluded sensitivity agrees NOT_CONFIRMED ($\delta_C = +0.2847$, one-sided 95% lower -0.0417; registered power 0.6085 at $n = 6$)	the Study-4 scoreboard section

Admission-regime difference (Studies 3 vs 4)	The v6 pools were built with the registered single-stratum admission screen active; all four v7 pools were built without it (the pool builder applies the screen only to versions v4–v6; the Study-4 registration is silent on the screen; identified at final review)	Internal / Construct	Arm-symmetric within Study 4, so it cannot bias $\Delta\delta$; disclosed with two sensitivities: S1 screened subset leaves 13 units (the 24-unit gate fails) at pooled share 0.0, so the graded signal is carried entirely by co-firing mutants that Study 3's regime excluded at admission; S2 PUT-level cluster bootstrap over the 15 distinct rich PUTs keeps the lower bound at $0.1818 > 0.15$, so the registered CONFIRM survives unit dependence	the Study-4 execution and scoreboard sections

Reviewer-reliability detail (Study 4). Expanding the reviewer-reliability row of the table above: the GP `bounds="fixed"` realization split independent blind reviewers 28 CONFIRMED to 22 REJECTED over 50 mutants; one same-content mutant pair received opposite labels; and the a3 anti-diffusive blow-up pattern drew opposite V2 judgements. All are recorded as reviewer-reliability data, with the mechanical AVP/equivalence pipeline authoritative for SMS. A post-hoc cross-vendor shadow review of 189 stratified frozen packets (`gpt-5.5`, `gemini-3.5-flash`; `review_shadow_kappa_v7.json`, seed 20260708) measured 92–93% agreement on random confirmations, 66–75% on rejections, and near-zero agreement on the contested `bounds="fixed"` family, where both text-only shadow vendors reject the instrumental-realization reading the execution-grounded reviewers accepted (shadow–shadow $\kappa = 0.80$; overall κ vs frozen 0.44/0.36). The cross-vendor consensus therefore sides with the stricter literal reading of that family; the frozen labels are unchanged, and sensitivity S1 already scores the stricter reading.

K.2 Semantic-Mutation Theory: Derivations and Proofs

This subsection carries the derivation prose and theorem proofs demoted from the main-text theory section (the main-text theory section); the theorem and proposition statements themselves remain in the main text.

Undecidability (the undecidability theorem), proof. S3 requires deciding $[\![P_e]\!]\not\models\psi$, a non-trivial semantic property, undecidable by Rice's theorem; the termination clause of S4 is independently undecidable by reduction from the halting problem. Semantic-mutant status therefore cannot be read off the edit; it can only be certified on bounded fragments with recorded budgets, which is the formal reason the admission judgement (the equivalence-judgement section, the prescreen section) is gate-shaped rather than a decision procedure. The theorem is a routine consequence of Rice's theorem, in the same spirit as the classical undecidability of mutant equivalence; we state it to fix the formal reason admission is gate-shaped, not as a novel result.

Fiber characterization (the fiber-characterization proposition), proof. (i) holds because semantics changes only through syntax; (ii) and (iii) follow by unfolding the definitions. The SMS-to-MS degeneration of the SMS-to-MS degeneration theorem (the degeneration section) is the metric-level shadow of (ii): when the admitted fiber collapses to the syntactic default, the SMS denominator collapses to the MS denominator.

Duality (the duality theorem), proof. If $P_e \equiv_\alpha P$ and r's violation predicate factors through α, then r flags P_e iff it flags P; MR validity says r does not flag P, so it does not flag P_e. The converse is the contrapositive. The duality identifies α as the single hinge connecting MR-side quality (strength) and mutant-side quality (activity). Classification is relative to $(I, \alpha, \text{budget})$: extending I can only move a mutant from active-off-taxonomy to ψ-stratified, never the reverse, and bounded $\equiv_\alpha$ verdicts are search conclusions, not proofs, so every audit record carries its $(I, \alpha, \text{budget})$ stamp.

Strong / weak MR and the strong boundary (full definitions). the duality theorem reads off strength algebraically, assuming the invariant ψ behind r is satisfied by the correct program so that closure under $\equiv_\alpha$ makes a violation diagnostic. For a discrete program P approximating a continuous operator this assumption is *contingent*: the correct P reproduces the operator's algebraic invariant only up to a discretisation residual, so $\equiv_\alpha$ between P and the ideal solution holds only within the tolerance ε_{tol} of the latency-window definition. Fixing ε_{tol}, call r *strong* on P when the correct P satisfies r within ε_{tol} (its residual stays below ε_{tol} under refinement), so that any violation beyond ε_{tol} certifies a semantically active mutant and the duality theorem holds operationally; call r *weak* on P when the correct P's residual exceeds ε_{tol}, so that the correct program is itself flagged because the discretisation, not a fault, breaks the invariant. The *strong boundary* is the $(\varepsilon_{\text{tol}}, \text{resolution})$ locus separating the two regimes; tightening ε_{tol} or coarsening the discretisation carries a strong MR across it into the weak regime. A weak MR is exactly a non-strong (non-grounded) MR in the effective, discrete sense: its violation set is not closed under the program's ε_{tol}-level $\equiv_\alpha$.

Boundary failures (the boundary proposition), proof. (i) and (ii) instantiate the Definition: weakness places the correct program's residual above ε_{tol}, and coincidental satisfaction places the mutant's signature below it; monotonicity of both sets in ε_{tol} holds because both membership predicates are threshold comparisons against the same ε_{tol}. Operationally, mode (i) does not inflate SMS: the killed predicate requires every counted MR to pass on the unmutated program, so a weak MR is rejected upstream at MR validation and contributes no kills. the boundary proposition(i) therefore manifests as MR-validation failure on the correct program, and only mode (ii) enters the SMS denominator-numerator arithmetic directly. The full reduction for the undecidability theorem follows Rice's theorem together with the standard halting reduction for the termination clause; the proof of the duality theorem is the $\equiv_\alpha$-closure argument above.

K.3 Higher-Order-Mutation Reachability, Full Analysis

The main text (the higher-order-mutation section) states the first-order scope of the unreachability claim and its conclusion; the full conceptual reachability analysis is reproduced here.

Conceptual analysis of HOM reachability. Higher-order mutation (HOM; 2, 3, 5) composes multiple first-order operators on the same source location. A natural challenge to this paper's preventive-defence framing is whether HOM compositions of cosmic-ray's 13 default operators or mutmut's 14 default operators can reach the the necessary-conditions section necessary conditions (a) cross-function-boundary, (b) domain-knowledge-dependent, or (c) algorithmic-class-change. We address this conceptually without empirical HOM testing.

For (a): HOM compositions remain over first-order operators that are AST-local within their target node; chaining AOR ∘ Statement-Deletion within a single function does not introduce a cross-module substitution like `det(M)` → `sum(np.diag(M))`, which presupposes that `np.diag` is in scope and semantically equivalent on diagonal matrices but not on general matrices. HOM can reach a kindred destination only if both operators happen to coincide with a meaningful cross-boundary swap, which is exponentially unlikely under the default operator menus.

For (b): HOM has no awareness of which constants are load-bearing in the PUT class. A two-operator chain such as Number-Replacer (1e-4 → 1e-3) ∘ Boolean-Swap on a Gaussian-process kernel does not "know" that 1e-4 is the regularization knob; it modifies the literal numerically without crossing the domain-semantics boundary. The Hyperparameter operator class (HP, mut_G) is by construction parameter-aware and class-specific, a property HOM compositions do not exhibit by default.

For (c): Algorithmic-class change (e.g. polynomial → spline basis; Lorenz RK4 → 1.5-order hybrid) requires structural code rewrites at the level of function bodies, library imports, or iteration schemes. HOM compositions over default first-order operators are bounded above by AST-local edits and so cannot reach algorithmic-class change without being effectively equivalent to a manual rewrite.

We therefore assert that the the AST-overlap section 0/0/0 unreachability for HP / SI / TF, while strictly empirical only for first-order configurations, is unlikely to be refuted by default-operator HOM compositions on combinatorial grounds. A direct empirical HOM falsification (generating mutmut second-order mutants and re-running the AST overlap analysis) remains a residual higher-order mutation threat.

A multi-syntactic-tool cross-comparison (mutmut and mutpy) was not run: mutpy is incompatible with Python 3.10+, and mutmut's operator set strongly overlaps cosmic-ray's. The full operator-class cross-reference between cosmic-ray (13 default operators) and mutmut (14 default operators), including which the necessary-conditions section necessary conditions each operator family reaches, is in Appendix B.6. The aggregate of the two default sets (~21 distinct first-order AST-local operator classes) collectively fails all three the necessary-conditions section necessary conditions.

Source distributional shift. The 15 cosmic-ray AST overlaps are not evenly distributed across the three LLMs (DeepSeek 11/15, Claude 4/15, GPT 0/15; see `cosmic_ray_12put_ast_diff.json`). DeepSeek tends to generate syntactically simpler mutations. This LLM-source bias is discussed as a distributional-shift threat (Appendix F.2) but does **not** affect the systematic-vs-incidental argument, which is based on *categorical* AST-locality, not hit frequency: the HP / SI / TF zero-overlap result is 0/0/0 across all three sources.

K.4 Study-1 Stipulated-Alternative Power Detail

The main text (the power-analysis section) states the H2 power verdict and its two headline figures; the full bootstrap exceedance and stipulated-alternative tables are reproduced here (the plug-in sample-size sweep itself is in Appendix D.3).

Table 25. Bootstrap exceedance probability under the observed distribution (plug-in, 5,000 replications, seed 42, over the unconditioned $n = 12/n = 48$ pool, $\delta = 0.439$).

Threshold	Interpretation	Exceedance prob. at (12, 48)
$\delta > 0.000$	Any effect	**0.997**
$\delta > 0.147$	Small	0.966
$\delta > 0.330$	Medium	0.759
$\delta >$ **0.474**	**Large (H2)**	**0.423**

The plug-in result answers "given the *observed* distribution, how often do we exceed the threshold?" A stipulated-alternative simulation answers a more pointed question: "if the *truth* equals the H2 boundary 0.474, how often does the (12, 48) design return $\delta \geq 0.474$?" SMS distributions have heavy ties at zero (45 of 60 cells), so a raw shift is discontinuous: any $\varepsilon > 0$ jumps δ from 0.314 to 0.74. We therefore use a mixture, drawing the aligned sample with probability w from (observed_aligned + 0.001) and with probability $1-w$ from observed_aligned, calibrating $w = 0.094$ to realise $E[\delta] = 0.4746$.

Table 26. Stipulated-alternative power at the H2 boundary.

Stipulated truth	Criterion	Power
0.474	$\hat{\delta} \geq 0.474$ (point estimate)	**0.491**
0.474	Lower 95% CI bound > 0 (any effect)	0.868

Even when the truth equals the H2 boundary, the point-estimate decision rule returns "not met" in roughly half of replications; a pass probability near one half at the boundary is intrinsic to a threshold rule applied to an approximately median-unbiased estimator at any sample size, so the 49.1% figure characterises the pre-registered decision rule, not a deficiency specific to the (12, 48) design, and its exact value depends on the stipulated alternative (the mixture with $E[\delta] = 0.4746$ is one of many alternatives with that mean). Increasing the sample size narrows the confidence interval but cannot lift the point estimate. The plug-in sample-size sweep (n_aligned ∈ {6, 12, ..., 60}; n_cross = 4 × n_aligned; power for $\delta > 0$ reaches 0.974 at n_aligned = 6 and 0.996 at 12, then plateaus) is in Appendix D.3.

K.5 Per-Class Operator Specialisation Catalogue

The main text (the coverage-matrix section) gives illustrative examples; the complete per-family specialisation catalogue is reproduced here (the full PUT-class × operator grid is in Appendix B.2).

- **mut_C Conservation-breaking:** A1 Lorenz adds $\varepsilon_{\text{drift}}$ to RHS (slow Hamiltonian drift); A2 LU decomposition omits the (k+1)-th row multiplier; B1 Beta-Bin posterior omits normalisation; C1 GPR covariance omits positive-definite diagonal term; D1 MLP backprop omits one gradient term.
- **mut_M Monotonicity-breaking:** A3 FDM Δt occasionally negative; B2 MCMC acceptance min(1, r) → min(0.95, r); C2 PCE high-order coefficient sort inserts inversion; D2 SVM decision-function sign flips near boundary.

- **mut_G Convergence-breaking:** A1 Lorenz RK4 → 1.5-order hybrid; A3 FDM 2nd-order difference → 1st-order; B3 MC doubling sample size does not 1/N-reduce variance; C3 NN-Surr training-epoch truncation.
- **mut_T Trajectory-distorting:** A1 Lorenz state-vector y / z swap; B2 MCMC inserts independent-sampling segment; C3 NN-Surr training-target slow phase shift; D1 MLP hidden-layer periodic-mask activation.
- **mut_F Fidelity-order-breaking:** A2 LU partial pivoting degrades to no pivoting; C1 GPR length-scale switches to coarse prior; C2 PCE high-order term randomly retains low-order; D3 LR regularisation occasionally large.

The per-class HP / OS / TF substitution rules give similarly differentiated specialisations across PUT classes (e.g., HP on class a = tolerance / max_iter, on class c = GPR `noise_level` / `length_scale`, on class d = MLP `hidden_dim` / dropout; OS on class a = numerical-linalg API swap `det` ↔ `sum(diag)`, on class b = sampling-API swap, on class c = surrogate-class swap GPR ↔ RBF ↔ NN).

K.6 Full PUT Selection Table

The main text (the program-selection section) gives a class summary; the full per-PUT table is reproduced here.

Table 27. PUT selection (full).

Class	PUT	Name	Mathematical structure	LOC
A Numeric	A1	Lorenz ODE integration	Nonlinear ODE	~150
	A2	LU decomposition	Linear algebra	~80
	A3	FDM 1D heat conduction	Parabolic PDE	~200
B Probabilistic	B1	Beta-Binomial conjugate	Analytic posterior	~60
	B2	MCMC Metropolis-Hastings	Markov chain	~250
	B3	Monte Carlo integration	Importance sampling	~100
C Surrogate	C1	Gaussian Process Regression	Kernel methods	~300
	C2	Polynomial Chaos Expansion	Orthogonal basis	~250
	C3	Neural-net surrogate	MLP substitution	~400
D ML	D1	Multi-Layer Perceptron	Backprop.	~350
	D2	Support Vector Machine	Convex optimisation	~200
	D3	Logistic Regression	Maximum likelihood	~120

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